Tutorial: Turing Machine Problems

1. Construct a Turing Machine for the language:

$$L = \{0^n 1^n 2^n \mid n \geq 1\}$$

2. Construct a Turing Machine for the language:

$$L = \{\alpha c \alpha \mid \alpha \in \{a, b\}^*\}$$

3. Construct a Turing Machine for the language:

$$L = \{a^i b^j c^k \mid i \times j = k,\ i, j, k \geq 1\}$$

4. Construct a Turing Machine for the language:

$$L = \{a^n b^m a^{n+m} \mid n, m \geq 1\}$$

5. Construct a Turing Machine for the language:

$$L = \{WW^R \mid W \in \{a, b\}^+\}$$

6. Construct a Turing Machine for the language:

$$L = \{0^{2^n} \mid n \geq 0\}$$

7. Design a Turing Machine to find the 1's complement of a binary number.

8. Design a Turing Machine to find the 2's complement of a binary number.